Tri-diagonal SVD

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1 Diagonalizing the Mass Matrix

We are deeply interested in diagonalising the mass matrix for the clockwork sfermions. This matrix is of the form:

$$M^2 = \begin{pmatrix} 1 & -\chi & & & \\ -\chi & 1+\chi^2 & -\chi & & \\ & \ddots & \ddots & \ddots & \\ & & -\chi & 1+\chi^2 & -\chi \\ & & & -\chi & \chi^2 \end{pmatrix} \tag{1}$$

where the 0 entries have been omitted to emphasise the tri-diagonal structure. To diagonalise the matrix, we require the eigenvectors of M^2, for an arbitrary gear number[1], N, i.e. $\dim(M^2) = N \times N$. To do so, we will consider the eigenvector as part of a sequence of N numbers, since this must be solved for an arbitrary number of dimensions, $N \in \mathbb{N}$. To do so, we will first delve into a few key ideas about sequences. This material is based on information gathered from [1].

1.1 The Ring of Sequences

Since the eigenvector is our primary, we will consider the sequence $v = \{0, v_1, \cdots, v_N, 0, 0, \cdots\}$. As such, this is noticably a Cauchy sequence (the elements get arbitrarily close together as the sequence progresses). Denote an arbitrary Cauchy sequence $f = \{f_m\}$, where it is understood that $m \in \mathbb{N}$ indexes over the Naturals. The k^{th} element of the sequence f is denoted by f_k.

We will equip the set of Cauchy sequences with two operations, "+" and "$*$", which will be referred to as "addition" and "convolution", respectively. Note that the "convolution" product will be realised as the convolution product in functional analysis, on the space of maps with elements $f : \mathbb{N} \to \mathbb{C}$, denoted by $l^{\mathbb{N}}$. This also aligns with the Cauchy product of a (formal) power series with the coefficients of each polynomial degree given by the corresponding element of the sequence. Furthermore, we will restrict ourselves, primarily,

[1] We will actually be using $N_x^i + 1$, but for the sake of brevity, will use N here instead.

to sequences with finite support (other than a few very specific examples).

$$(f+g)_k = f_k + g_k \tag{2}$$

$$(f*g)_k = \sum_{m=0}^{k} f_m g_{k-m} \tag{3}$$

Note that the "+" on the right hand side of equation (2) is the regular addition over the field $\mathbb{C}$, since $f_k, g_k \in \mathbb{C}$. Clearly, the additive identity over this space is given by $\mathrm{id}^{+}_{l^{\mathbb{N}}} = \{0, \cdots\}$, the sequence of additive identities over $\mathbb{C}$. We will make judicious use of specialised notation to denote important sequences:

$$\bar{c} = \{c\,\delta_0^m\}, \quad c \in \mathbb{C} \tag{4}$$

$$\delta_n = \{\delta_n^m\} \tag{5}$$

$$\Theta_n = \{\sum_{j=0}^{m} \delta_j^n\} \tag{6}$$

Equation (4) presents the scalar sequence[2], eq. (5) the delta sequence[3], and eq. (6), the Heaviside sequence, with Θ_n having vanishing entries up to n (non-inclusive), and 1 thereafter.

Under these two operations, the space of such sequences, $l^{\mathbb{N}}$ forms a commutative ring[4] with no zero divisor, i.e. $g*f = \bar{0} \implies g = \bar{0}$ or $f = \bar{0}$, where $\bar{0}$ is the additive identity, $\mathrm{id}^{+}_{l^{\mathbb{N}}}$. Let us prove this:

Lemma 1. *For $f, g \in l^{\mathbb{N}}$, $f*g = \bar{0} \implies (g = \bar{0})$ or $(f = \bar{0})$.*

Proof. We will prove the contra-positive: $(g \neq \bar{0})$ and $(f \neq \bar{0}) \implies f*g \neq \bar{0}$.

Assume that $f \neq \bar{0}$ and $g \neq \bar{0}$. This implies that the sets of non-zero indices for each sequence, $I_f = \{i \in \mathbb{N} \,|\, f_i \neq 0\}$ and $I_g = \{i \in \mathbb{N} \,|\, g_i \neq 0\}$, are non-empty. Let $\alpha = \min I_f$ and $\beta = \min I_g$ be the indices of the first non-zero elements in each sequence, with their existence guaranteed by the well-ordering principle.

Observe the action of δ_t on an arbitrary sequence h, via convolution. We will freely use the commutativity of the convolution.

$$\delta_t * h = \{\sum_{j=0}^{m} \delta^t_{m-j} h_j\} = \{h_{m-t}\Theta(m-t)\}$$

where $\Theta(m-t)$ is the discrete Heaviside step function that vanishes if $m < t$ and evaluates to 1 otherwise. Thus, convolution with δ_t shifts all entries in the sequence forward by t steps.

[2]This will become clear as we use its convolution properties.

[3]If interpreted in the sense of a distribution, δ_n is the shifted (by n) Dirac distribution.

[4]The commutative, distributive and associative properties are easy to prove with the convolution formula, and will be omitted.

Then, there are sequences[5] $F, G \in l^{\mathbb{N}}$ such that $\delta_\alpha * F = f$ and $\delta_\beta * G = g$. The first entry in F and in G are non-zero, since they equal[6] f_α and g_β. The associativity and commutativity of the convolution operation imply:

$$f * g = (\delta_\alpha * F) * (\delta_\beta * G) = (\delta_\alpha * \delta_\beta) * (F * G)$$

We note that $\delta_\alpha * \delta_\beta = \delta_{\alpha+\beta}$ by replacing $h = \delta_\beta$ above. A convolution with $\delta_{\alpha+\beta}$ would move the position of every sequence entry by $\alpha + \beta$, with the first $\alpha + \beta$ entries being 0. Observe that for $x, y \in l^{\mathbb{N}}$, we have $x * y = \{x_0 y_0,\, x_1 y_0 + x_0 y_1,\, \cdots\}$. Since $F_0 \neq 0$ and $G_0 \neq 0$, $F * G$ is a non-vanishing sequence, by virtue of its first entry. As a result, $(\delta_{\alpha+\beta} * F * G)_{\alpha+\beta} = (f * g)_{\alpha+\beta} \neq 0$, concluding the proof of the contra-positive. This is sufficient to conclude that $f * g = \bar{0} \implies (g = \bar{0})$ or $(f = \bar{0})$. □

Thus, $(l^{\mathbb{N}}, +, *)$ forms a commutative ring[7] with no non-trivial zero divisors, implying that it is an integral domain[8]. However, it is not a field, as one may verify by attempting to solve $\delta_1 * x = f$ for some sequence x, since $(\delta_1 * x)_0 = 0$, which requires that $f_0 = 0$. Consider the restriction to a subspace of sequences $G_{l^{\mathbb{N}}} = \{h \in l^{\mathbb{N}} \mid h_0 \neq 0\}$. This space is not closed under addition, but $(G_{l^{\mathbb{N}}}, *)$ forms a group.

Theorem 2. *Given $f \in G_{l^{\mathbb{N}}}$ and $g \in l^{\mathbb{N}}$, one can find a unique $x \in l^{\mathbb{N}}$, such that $f * x = g$.*

Proof. A proof by induction is in order. Note that $f_0 \neq 0$ by definition, and that $(f * x)_0 = f_0 x_0$. If we were to consider the equation $f * x = g$, we may establish $x_0 = \frac{g_0}{f_0}$. At this step, it is easy to see that if g is restricted to $G_{l^{\mathbb{N}}}$, so is x, since $g_0 \neq 0 \implies x_0 \neq 0$.

Let $k \in \mathbb{N}$ and assume that we have been able to solve for the first $k + 1$ elements of x, $\{x_0, \cdots, x_k\}$. Then,

$$\begin{aligned} g_{k+1} &= (f * x)_{k+1} \\ &= \sum_{j=0}^{k+1} f_j x_{k-j+1} \\ &= f_0 x_{k+1} + \sum_{j=1}^{k+1} f_j x_{k-j+1} \end{aligned}$$

Clearly, knowledge of the first $k + 1$ elements of x allows us to determine x_{k+1}, as long as $f_0 \neq 0$ (which is satisfied by the fact that $f \in G_{l^{\mathbb{N}}}$).

Since we are able to solve for x_0, we must be able to solve for x_1, x_2, etc. The principle of mathematical induction allows us to conclude our proof. □

Remark 3. *The statement that restricting g to implies $x \in$ is quite powerful. In fact, this gives additional structure.*

[5] F, G are simply f, g with all the leading 0 entries "deleted".

[6] One can simply replace ($h = F$, $t = \alpha$) and ($h = G$, $t = \beta$) and set the resultant sequence to f and g respectively.

[7] The convolution identity being δ_0.

[8] Cheng claims that one may construct a field of quotients from $l^{\mathbb{N}}$. However, this is untrue, and some care must be taken to avoid defining inverses for singular sequences.

Corollary 4. $(G_{l^{\mathbb{N}}}, *)$ *forms an abelian group.*

Proof. The associativity and commutativity follow from the properties of the convolution, and can be verified explicitly by equation (3). The fact that $(f * g)_0 = f_0 g_0$ implies the closure of $(G_{l^{\mathbb{N}}}, *)$. The exploration of convolution with δ_t in Lemma 1 establishes that δ_0 is the group identity, "translating" elements in the sequence by 0 units. Clearly, $\delta_0 \in G_{l^{\mathbb{N}}}$. Finally, an inverse is required of every element. The existence of an inverse is guaranteed by Theorem 2, where we may set $g = \delta_0$ and provide x the additional moniker of f^{-1}. □

1.2 A Few Properties

We will look at a few properties of sequences under convolution.

$$\bar{c} * f = \{cf_m\} = cf, \qquad c \in \mathbb{C}, \qquad \text{Scalar Convolution} \quad (7)$$
$$\delta_k * f = \{f_{m-k}\Theta(m-k)\}, \qquad \text{Translation} \quad (8)$$
$$\delta_0 * f = f, \qquad \text{Identity} \quad (9)$$
$$(\bar{a} - \delta_1)^{-1} = \{a^{-(m+1)}\}, \qquad a \in \mathbb{C} - \{0\} \qquad \text{Geometric Inverse} \quad (10)$$

The inverse of the $(\bar{a}-\delta_1)$ sequence should be quite reminiscent of the Taylor series expansion for $(1-x)^{-1}$, the Geometric series. Since $a \neq 0$ is assumed, $(\bar{a} - \delta_1) \in$, which implies that it has an inverse. Then, we may verify that it is indeed the inverse by simply checking that $(\bar{a}-\delta_1)*\{a^{-(m+1)}\} = \delta_0$. The Scalar Convolution identity implies that $\bar{a}*\{a^{-(m+1)}\} = \{a^{-m}\}$. Furthermore, the Translation identity implies that $\delta_1 * \{a^{-(m+1)}\} = \{\Theta(m-1)\, a^{-m}\}$. Thus, $(\bar{a} - \delta_1) * \{a^{-(m+1)}\} = \delta_0$, which establishes their inverse relation.

1.3 Sequential Solutions

Bearing the notion of "sequence algebra" in mind, we may proceed to derive the eigenvectors and eigenvalues of the mass matrix (1). Let an arbitrary eigenvector be denoted by V, an N dimensional vector, and v, the sequence containing it: $v = \{0, V_1, \cdots, V_N, 0, 0, \cdots\}$. Let λ be the corresponding eigenvalue. Then, $M^2V = \lambda V$ can be expressed as a set of conditions on the sequence v. Barring the first 2 entries and the $(N+1)^{th}$, the eigenvalue relation can be expressed as:

$$-\chi(v_{k-1} + v_{k+1}) + (1+\chi^2)v_k = \lambda v_k$$

Let us define another sequence containing the "boundary terms", $b = \{\chi^2 v_1 \delta_1^m + v_N \delta_N^m\}$. These are the terms that prevent the mass matrix (1) from being Toeplitz. We will now use this to define the full eigenvector relation:

$$-\chi\{v_{m+2}\} + (1+\chi^2)\{v_{m+1}\} - \chi\{v_m\} = \lambda\{v_{m+1}\} + \{b_{m+1}\} \qquad (11)$$

We may proceed by noting that $\delta_1 * \{g_{m+1}\} = g - \bar{g}_0$, and $\delta_2 * \{g_{m+2}\} = g - g_1\delta_1 - \bar{g}_0$. Additionally, we have $\delta_2 = \delta_1 * \delta_1 \equiv (\delta_1)^2$. We may operate on equation (11) with $(\delta_1)^2$, to obtain:

$$-\chi(v - \bar{v}_0 - v_1\delta_1) + (1 + \chi^2 - \lambda)\delta_1 * (v - \bar{v}_0) - \chi(\delta_1)^2 * v = \delta_1 * (b - \bar{b}_0)$$

Using the fact that $v_0 = b_0 = 0$, and denoting $\chi\Delta = 1 + \chi^2 - \lambda$ we may simplify this to obtain:

$$(-\chi(\delta_1)^2 + \chi\Delta\delta_1 - \bar{\chi}) * v = \delta_1 * (b - \chi\bar{v}_1) \tag{12}$$

Note that on the left side, the sequence that is being convolved with v has a term proportional to δ_1, a term proportional to δ_2 and a scalar sequence, $\bar{\chi}$. As a result, as long as $\chi \neq 0$ (which implies that there is no CW), we may find an inverse to the sequence, and solve for v, with the solution:

$$v = \frac{1}{\chi}\frac{\delta_1 * (\chi\bar{v}_1 - b)}{((\delta_1)^2 - \Delta\delta_1 + \bar{1})}, \tag{13}$$

where the quotient is understood to indicate the convolution of the numerator with the inverse of the denominator. Since $l^{\mathbb{N}}$ is a commutative ring, the polynomial denominator can be decomposed into a product $\chi(\bar{a} - \delta_1)(\bar{b} + \delta_1)$. Furthermore, since $\chi \neq 0$, $ab \neq 0$, implying that both of the decomposed sequences are invertible. As a result, we may carry out the procedure of partial fraction decomposition, treating the restricted sequences as elements of a field.

$$\frac{1}{((\delta_1)^2 - \Delta\delta_1 - \bar{1})} = \frac{1}{(a_+ - a_-)}\frac{1}{\bar{a}_- - \delta_1} - \frac{1}{\bar{a}_+ - \delta_1}$$

From this, we have the factorisation conditions: $\bar{a}_+ * \bar{a}_- = \bar{1} \implies a_+a_- = 1$, and $a_+ + a_- = \Delta$. This can be solved exactly, with the solutions:

$$a_\pm = \frac{1}{2}(\Delta \pm \sqrt{\Delta^2 - 4}) = re^{\pm\theta}$$

We have $r = 1$ by virtue of $a_+a_- = 1$, and $\cos\theta = \frac{1}{2\chi}(1 + \chi^2 - \lambda)$. We must now address 2 cases. Either $a_+ = a_-$, or $a_+ \neq a_-$.

1.3.1 Case 1: $a_+ = a_-$

In this case, $\Delta = \pm 2$, since $a_+ - a_- = 0 = \sqrt{\Delta^2 - 4}$. In this case, we know that $\lambda = (1 + \chi^2) \pm 2\chi = (\chi \pm 1)^2$. Then, we may factor the polynomial in δ_1 as an exact square:

$$v = \frac{\delta_1 * (\chi\bar{v}_1 - b)}{\chi(\pm\bar{1} - \delta_1)^2} \tag{14}$$

Making use of the Geometric Inverse identity from subsection 1.2 once again, we find that

$$\begin{aligned} v &= \frac{1}{\chi}\{(\pm 1)^{-(m+1)}\} * \{(\pm 1)^{-(m+1)}\} * \delta_1 * (\chi\bar{v}_1 - b) \\ &= \frac{1}{\chi}\{(\pm 1)^{-(m+1)}\} * \{(\pm 1)^{-m}\Theta(m-1)\} * (\chi\bar{v}_1 - b) \\ &= \frac{1}{\chi}\{(\pm 1)^{-(m+1)}m\} * (\chi\bar{v}_1 - b) \end{aligned}$$

We can now form an explicit expression for the sequence v, noting that $(\pm 1)^{-(m+1)} = (\pm 1)^{m+1}$;

$$v_m = \frac{1}{\chi}\chi v_1(\pm 1)^{m+1}m - \chi^2 v_1(\pm 1)^m(m-1)\Theta(m-1) - v_N(\pm 1)^{m-N+1}(m-N)\Theta(m-N) \tag{15}$$

Let us first eliminate v_N:

$$v_N = v_1(\pm 1)^N - \chi(N-1) \pm N$$

We may further impose the condition that $v_{N+1} = 0$, leading us to the restriction:

$$\frac{Nv_1}{\chi}(\pm 1)^N 2\chi \mp \chi^2 \mp 1 = -\frac{Nv_1}{\chi}(\pm 1)^{N+1}(\chi \mp 1)^2 = 0$$

$v_1 \neq 0$ must be established to prevent the sequence v from vanishing identically, since v_N, and thus v, are proportional to v_1. Avoiding this would provide the 0 vector. Then, this case is only useful to us if $\chi = \pm 1$, which would not produce the CW mechanism. As such, we discard the possibility of $a_+ = a_-$.

1.3.2 Case 2: $a_+ \neq a_-$

We make use of the Geometric Inverse identity from subsection 1.2, to note that:

$$\frac{1}{\bar{a}_\pm - \delta_1} = \{a_\pm^{-(m+1)}\} = \{e^{\mp(m+1)\theta}\}$$

Further, note that $a_+ - a_- = 2\sin\theta$. We may plug all of this back into equation (13) to obtain the desired sequence.

$$\begin{aligned} v &= \frac{1}{\chi(a_+ - a_-)}\frac{1}{\bar{a}_- - \delta_1} - \frac{1}{\bar{a}_+ - \delta_1} * \delta_1 * (\chi\bar{v}_1 - b) \\ &= \frac{1}{\chi\sin\theta}\{\sin((m+1)\theta)\} * \delta_1 * (\chi\bar{v}_1 - b) \\ &= \frac{1}{\chi\sin\theta}\{\sin(m\theta)\Theta(m-1)\} * (\chi\bar{v}_1 - b) \end{aligned}$$

Since $\sin(0\theta) = 0$, we may discard the Θ function. Then, we may expand this to find the sequence element:

$$v_m = \frac{1}{\chi\sin\theta}\chi v_1 \sin(m\theta) - \chi^2 v_1 \sin((m-1)\theta)\Theta(m-1) - v_N \sin((m-N)\theta)\Theta(m-N)$$

Setting $m = N$ allows us to determine v in terms of v_1.

$$v_N = \frac{v_1}{\sin\theta}\sin(N\theta) - \chi\sin((N-1)\theta)$$

Thus, we obtain our final eigenvector equation, given an eigenvalue:

$$\begin{aligned} v_m = \frac{v_1}{\chi\sin\theta}\Big(&\chi\sin(m\theta) - \chi^2\sin((m-1)\theta)\Theta(m-1) \\ &- \frac{1}{\sin\theta}\sin(N\theta) - \chi\sin((N-1)\theta)\sin((m-N)\theta)\Theta(m-N)\Big) \end{aligned} \tag{16}$$

We may now proceed to employ the boundary condition of $v_{N+1} = 0$. This will place constraints on the available values of θ, and as a result, the eigenvalue. it will be helpful to recall that $\cos\theta = \frac{\Delta}{2} = \frac{1}{2\chi}(1+\chi^2-\lambda)$.

$$\frac{v_1 \sin(N\theta)}{\chi \sin\theta}\big(2\chi\cos(\theta) - (1+\chi^2)\big) = 0, \tag{17}$$

where the trigonometric identity $\sin\left((N+1)\theta\right) + \sin\left((N-1)\theta\right) = 2\cos(\theta)\sin(N\theta)$. By defining θ as the complex (principal) argument, we have restricted it to $(-\pi, \pi] \subset \mathbb{R}$. Thus, $|\cos(\theta)|$ is bounded by 1. To have $2\chi\cos(\theta) - (1+\chi^2) = 0$, we require $(\chi-1)^2 \leq 0$ if $\chi > 0$, or $(\chi+1)^2 \leq 0$ if $\chi < 0$. These cases restrict us to $\chi = \pm 1$, which provides no CW suppression at all. Thus, we must have $\sin(N\theta)$ vanish, since v_1 returns the trivial sequence. Clearly, this restricts us to $\theta_k = \frac{k\pi}{N}$, giving us our N eigenvalues:

$$\lambda_k = \chi^2 - 2\chi\cos\frac{k\pi}{N} + 1, \qquad k \in \{1, \ldots, N-1\} \tag{18}$$

It is also useful to introduce the eigenvector corresponding to the k^{th} eigenvalue, valid for $k > 0$:

$$v_m^{(k)} = \frac{v_1^{(k)}}{\sin\frac{k\pi}{N}}\left(\sin\frac{mk\pi}{N} - \chi\sin(m-1)\frac{k\pi}{N}\Theta(m-1)\right), \qquad k \in \{1, \cdots N-1\} \tag{19}$$

For $v^{(N)}$, recall the discussion of the zero-mode in section **??**, normalised:

$$v_m^{(N)} = \sqrt{\frac{\chi^2-1}{\chi^2-\chi^{2(1-N)}}}\,\chi^{1-m}, \qquad \lambda_N = 0 \tag{20}$$

All that remains in the diagonalisation procedure is to normalise these eigenvectors, so that they may be interpreted as the columns of a rotation matrix.

$$\sum_{m=1}^{N}(v_m^{(k)})^2 = \frac{N(v_1^{(k)})^2}{2\sin^2\frac{k\pi}{N}}(\chi^2 - 2\chi\cos\frac{k\pi}{N} + 1) = \frac{N(v_1^{(k)})^2}{2\sin^2\frac{k\pi}{N}}\lambda_k$$

We may now completely specify our diagonalising matrix:

$$R_{jk} = \sqrt{\frac{2}{N\lambda_k}}\left(\sin\frac{jk\pi}{N} - \chi\sin(j-1)\frac{k\pi}{N}\Theta(j-1)\right) + \delta_k^N\sqrt{\frac{\chi^2-1}{\chi^2-\chi^{2(1-N)}}}\,\chi^{1-j} \tag{21}$$

For the sake of notational convenience, we assume that the term that is not proportional to δ_N^k vanishes as $k \to N$, despite $\lambda_N = 0$. Since M^2 is a real symmetric matrix, we are guaranteed orthogonal eigenvectors, implying $R^T R = \mathbb{I}$. Thus, we have that $m^2 = R^T M^2 R$ is a diagonal matrix with eigenvalues given by λ_k.

References

[1] S. S. Cheng. *Partial Difference Equations.* CRC PRESS, 2003.